

Q-Framework Evidence Whitepaper v1.0

RC3

Verifiable Heterogeneous AI Compute Continuation

Public snapshot: October 7, 2026

Q-Framework is a running proprietary research architecture for heterogeneous AI compute. Its central question is not whether several GPUs can be described as one larger physical GPU. The deeper question is:

Must the lifecycle of a running AI workload remain permanently bound to one GPU, one process, or one node?

RC3 publishes real validation evidence while intentionally keeping the production mechanism black-box.

1. Publication principle

Real evidence. Black-box mechanism.

Measured results, hashes, comparison metrics, completed outputs and sanitized hardware classes are real development records. Production state schemas, algorithms, equations, coefficients, restore sequencing, scheduler policy and source code remain proprietary.

2. The research problem

Q-Framework explores whether completed computation can remain useful after resource pressure or execution interruption; whether workloads can continue across heterogeneous workers; whether independent workers can share a verifiable reference authority; and whether long-running work can eventually be intentionally persisted and resumed.

3. This is not a monitoring system: OOM control is part of the execution path

The public **Live Compute** page is an observation and evidence surface. It displays GPU / VRAM state, QmRNA, QSTATE, branch progress, node heartbeat and runtime progress, but **telemetry is not the core Q-Framework mechanism**.

The production path has been exercised inside real MiniMax H3 / ComfyUI-class workloads. At a high level, validated behavior includes:

- creating recoverable checkpoint / QSTATE / QREV state under VRAM pressure / PRE-OOM conditions;
- spilling retained computation state to CPU RAM or SSD-backed storage rather than merely logging memory usage;
- releasing GPU residency to recover usable VRAM;
- restoring saved state to the same GPU for **same-GPU continuation**;
- or restoring it to another GPU worker for **mixed-GPU continuation / rescue**;
- continuing sampler / model execution after restore instead of restarting the entire workload;
- verifying continuation lineage with receipts, SHA and numerical evidence.

The engineering objective is therefore not to "observe OOM." It is to change the workload lifecycle around memory pressure: **preserve useful computation before failure where possible, release residency, restore state, and continue execution**.

This differs from a VRAM monitor/debug node. It is also not equivalent to merely calling `empty_cache()`, reducing resolution, quantizing a model, or lowering an attention chunk size. Those techniques can be compatible local optimizations; Q-Framework addresses a higher-level problem: **stateful memory-pressure recovery plus resumable execution**.

Public evidence already includes real H3 continuation, completed output after cross-GPU restore, bitwise integrity of an approximately 250 MiB transported state object, and multi-worker execution lineage. Production pressure scoring, state packing, residency/eviction policy, restore sequencing and model-forward modifications remain proprietary.

Q-Framework is an OOM-aware execution / recovery implementation, not an OOM telemetry dashboard.

Claim discipline still applies: this does not mean "every model can never OOM" or "zero OOM probability." The supported claim is that **Q-Framework has implemented and validated execution paths that can preserve, release, restore and continue real AI workload state under VRAM-pressure and OOM-recovery scenarios**.

Name clarification

This project is not affiliated with Unity QFramework, NVIDIA CUDA-Q, or other similarly named frameworks. In this repository, **Q-Framework** refers specifically to the heterogeneous AI state-virtualization, OOM-aware recovery and compute-continuation architecture documented here.

4. Four validated capability families

Mixed-GPU Compute Continuation

A controlled 20-step continuation test produced bitwise equality, allclose=true, max absolute difference 0.0 and SHA-256:

```
ae04c23d4daed035632855260f85c360a857c28a8d46544f5c92f91baff823a3
```

Same-Worker / Same-GPU Continuation

Continuation has also been exercised without unnecessary migration when locality is preferable. Production checkpoint structure and resume policy are not disclosed.

H3 Memory Static / Shared Reference-State Compute

A cross-node reference-state validation produced the same SHA-256:

```
1488ccf08323200643d5759046f2d47e1e16d44a70f278a79346aec53103d937
```

Mixed-GPU Cross-Boundary Compute

Execution lineage has crossed GPU-worker, node and media/artifact boundaries. Recorded output hashes include:

```
7A909C59636C1A18C6B1A1C7CD2037120F25DDC2C7F36CD2E62B4C70A43212E1
```

```
C35FAECFE3C1BD0B5338CC4BE0EB8A854A10CE0FEEF86D39CA750DE6BB4A357C
```

5. Large-state transport integrity

A ~250 MiB frozen computation object preserved exact SHA through the tested transport path.

SHA-256:

```
330674e23bf109a4f5e3983608ce118274373deae669e3d3bacee2f77639d5a6
```

Bitwise SHA match was true and measured end-to-end test time was approximately 3.675 seconds.

6. From controlled proof to real H3 workload

A real H3 workload completed after continuation on another GPU worker.

- 480 x 832
- 24 fps
- 39 frames
- output SHA-256: 307828DE5D0D54F36004EF9C97B2E4DAE5178E59A9C8EF74775F07522B788AB3

7. Four independent 22 GB GPU workers

A production-class test used four independent RTX 2080 Ti 22 GB workers within one execution lineage.

- duration: 15.000 s
- frames: 540
- participating workers: 4
- final SHA-256: 23CDBA13DBFF7B8EC234DB1D7DE61625597E128BD6684849FCB6CD33650ADF57

This is not a claim that four 22 GB GPUs physically become one 88 GB GPU.

8. Cross-environment numerical consistency

Three execution environments independently produced the same recomputed-result SHA:

330271d14ee7086e27f875a4a56e720f169cdef1523e908ad94696bb70df15a4

Float32 reconstruction comparison against source:

- max absolute error: 4.76837158203125e-07
- mean absolute error: approximately 2.03e-08

Exact hashes and numerical tolerance are deliberately reported as different forms of evidence.

9. QSTATIC + QmRNA controlled equivalence

A controlled A/B test used the same reference, seed and generation conditions. MP4 container hashes differed, so decoded visual content was compared.

- decoded frames: 22
- exact matching frame hashes: 22 / 22
- framemd5 manifest SHA-256:

```
FF55120ADF80F99D74405C5DECA7A9B54116B01F987591A012BD1FF01467F5D7
```

The internal QmRNA representation, QSTATIC contracts and production control path remain proprietary.

10. Why the evidence matters

Q-Framework separates transport integrity, computed-result consistency and decoded-output equivalence. This avoids treating “looks similar” as equivalent to cryptographically or numerically verifiable consistency.

11. The larger hypothesis

Q-Framework is testing a different systems model:

The GPU can be an execution worker without being the permanent owner of the workload lifecycle.

If this continues to hold under harder conditions, it may have implications for consumer-GPU utilization, mixed-generation fleets, edge/workstation AI, interrupted-workload recovery, long-running generative workloads and heterogeneous compute economics.

These are research directions, not unbenchmarked performance promises.

12. Subsequent validation stages

RC3 is not an endpoint. Q-Framework development is not framed as a binary success/failure story.

Every solved engineering problem creates a harder test.

Current validation priorities include:

- longer continuous workloads;
- wider GPU generation / driver / runtime combinations;

- intentional pause -> persist -> later resume;
- lower wasted recomputation and shorter time-to-result;
- repeatability across more runs, nodes and hardware.

13. Public progress and future milestones

Future public milestones may include longer completed outputs, new SHA receipts, repeated-run statistics, broader hardware matrices, pause/persist/resume evidence, worker-takeover evidence, time-to-result comparisons and new QSTATIC/QmRNA controlled-equivalence results.

This whitepaper is intended to remain a continuously updated public technical record. Future revisions will incorporate additional measurements, validation results and evidence as they become suitable for disclosure.

14. Experimental Preview

Q-Framework plans a limited **Experimental Preview** to allow external researchers, developers and prospective collaborators to evaluate selected validated capabilities under controlled conditions without receiving the complete production engine.

Possible preview surfaces include limited continuation experiments, pause/persist/resume demonstrations, receipt verification, reference-state consistency tests and selected heterogeneous-worker demonstrations.

The Preview is intended to be packaged and constrained. Proprietary production mechanisms remain black-box.

Future preview releases may use a staged build model:

Experimental Preview — Build #001 / #002 / #003 ...

15. Disclosure boundary

Public: measured outcomes, hashes, comparison metrics, sanitized hardware/workload metadata, completed-artifact evidence and milestone progress.

Private: production state schemas, serialization, restore sequencing, QmRNA representation/equations, QSTATIC contracts, Qvram residency, pressure thresholds, eviction/prefetch, scheduler/worker-selection, lease/fencing/retry implementation, production model-forward modifications, private topology and source code.

Black-box does not mean false evidence.

| The evidence must be real. The implementation does not have to be given away.

16. Claim discipline

Q-Framework does not currently claim unlimited VRAM, zero OOM probability, universal model compatibility, universal bitwise determinism, physical VRAM aggregation, or universal acceleration without benchmark support.

Claims follow evidence.

17. Conclusion

The emerging evidence chain is:

verifiable state -> result-preserving continuation -> real H3 continuation -> multi-GPU participation -> cross-environment consistency -> cross-node reference consistency -> controlled runtime-path output equivalence.

The next objective is to push this chain toward longer, faster, more heterogeneous and more repeatable workloads.

The production mechanism remains black-box.

Real evidence. Black-box mechanism. Future progress will be reported through verifiable results.